In this, SVM and Decision Tree regression models are implemented. Different parameters are changed and their R^2 scores are compared using tables. This helps to identify the best parameters and understand how model performance varies.

Support Vector Machine

S.No	Regularization parameter, C	**Linear** ***r_score***	**Poly** ***r_score***	**Rbf** ***r_score***	**Sigmoid** ***r_score***	**Gamma**
1.	1.0	-0.055	-0.057	-0.057	-0.057	Auto
2.	1.0	-0.055	-0.057	-0.057	-0.055	Scale
3.	100	0.106	-0.019	-0.050	-0.030	Auto
4.	100	0.106	-0.019	-0.050	-0.030	Scale
5.	1000	0.780	0.266	0.006	0.185	Auto
6.	1000	0.780	0.266	0.006	0.185	Scale
7.	2000	0.876	0.481	0.067	0.397	Auto
8.	2000	0.876	0.481	0.067	0.379	Scale
9.	3000	0.895	0.637	0.123	0.591	Auto
10.	3000	0.895	0.637	0.123	0.591	Scale

SVM: Increasing C reduced under fitting and improved model performance. The ***Linear kernel*** with ***C=3000*** gave the best result.

Decision Tree

S.No	Criterion	Splitter	Max_features	r_score
1.	Squared error	best	none	0.892
2.	Squared error	random	none	0.882
3.	Squared error	best	sqrt	0.442
4.	Squared error	random	sqrt	0.598
5.	Squared error	best	log2	0.548
6.	Squared error	random	log2	0.733
7.	Absolute error	best	none	0.951
8.	Absolute error	random	none	0.827
9.	Absolute error	best	sqrt	0.442
10.	Absolute error	random	sqrt	0.681
11.	Absolute error	best	log2	0.909
12.	Absolute error	random	log2	0.395
13.	Friedman mse	best	none	0.905
14.	Friedman mse	random	none	0.919
15.	Friedman mse	best	sqrt	0.464
16.	Friedman mse	random	sqrt	0.173
17.	Friedman mse	best	log2	-0.773
18.	Friedman mse	random	log2	0.631

The best performance was achieved using '***absolute error'* criterion with '*best'* splitter and Max_features='*None*'**, giving an R^2 score of **0.951**.